

Multiple Choice (10 marks)

1. Use a number line to find the sum of $-9 + (-8)$.
 - (a) -17
 - (b) 1
 - (c) -1
 - (d) 17
2. Jonathan obtained a score of 80 on a statistics exam, placing him at the 90 th percentile. Suppose five points are added to everyone's score. Jonathan's new score will be at the
 - (a) 80 th percentile.
 - (b) 85 th percentile.
 - (c) 90 th percentile.
 - (d) 95 th percentile.
3. Find the volume of the solid generated when the region bounded by the y-axis, $y = e^x$, and $y = 2$ is rotated around the y-axis.
 - (a) 0.296
 - (b) 0.592
 - (c) 2.427
 - (d) 3.998
4. How many positive cubes divide $3! \cdot 5! \cdot 7!$?
 - (a) 6
 - (b) 4
 - (c) 3
 - (d) 1
5. Two whole numbers have a least common multiple of 60. Each number is less than or equal to 12. The greatest common factor of the two numbers is 2. What are the two numbers?
 - (a) 6 and 10
 - (b) 5 and 12
 - (c) 10 and 12
 - (d) 12 and 15

True / False (10 marks)

Answer True or False

6. True or False: The upper critical value of z for a 94% confidence interval is 1.88.
7. True or False: The difference in elevation between Talon Bluff and Salt Flats is 565 feet.
8. True or False: -3 is greater than -5.
9. True or False: The solution to the equation $3/2 * w = 66$ is $w = 44$.
10. True or False: The simplified expression of $(9x^9 + 7x^8 + 4x^7) + (x^{11} + x^9 + 2x^7 + 3x^3 + 5x + 8)$ is $x^{11} + 10x^9 + 7x^8 + 6x^7 + 3x^3 + 5x + 8$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the concept of residuals in regression analysis. Why is it significant that the mean of the residuals is always zero, and how does this impact the accuracy of the regression model?
12. In the first quadrant, analyze the polar curve $r = \sin 3$ to determine the angle at which the curve has a vertical tangent. Explain your reasoning and calculations.

End of Paper